Capstone Project Virtual Showcase

Summer 2026 — July 31st, 2026

FIU Knight Foundation School of Computing and Information Sciences

Fitness & Wellness Tracker

Fitness & Wellness Tracker Team – Computer Science

The Problem

- Many students and busy adults want to stay consistent with workouts, meals, and wellness goals, but their progress is split across notes, spreadsheets, or separate apps that do not work together.
- Without one place to log exercise and nutrition, it is hard to see whether daily habits support longer-term goals.
- Existing fitness apps can feel heavy, social-first, or device-dependent, which is more than we need for a focused semester MVP.
- Generic trackers often leave it unclear how frontend screens, API rules, and stored user data should connect in a simple web stack.
- There is a gap for a student-built, web-only tracker with clear architecture, private accounts, and a realistic roadmap from planning into implementation.

Our Solution

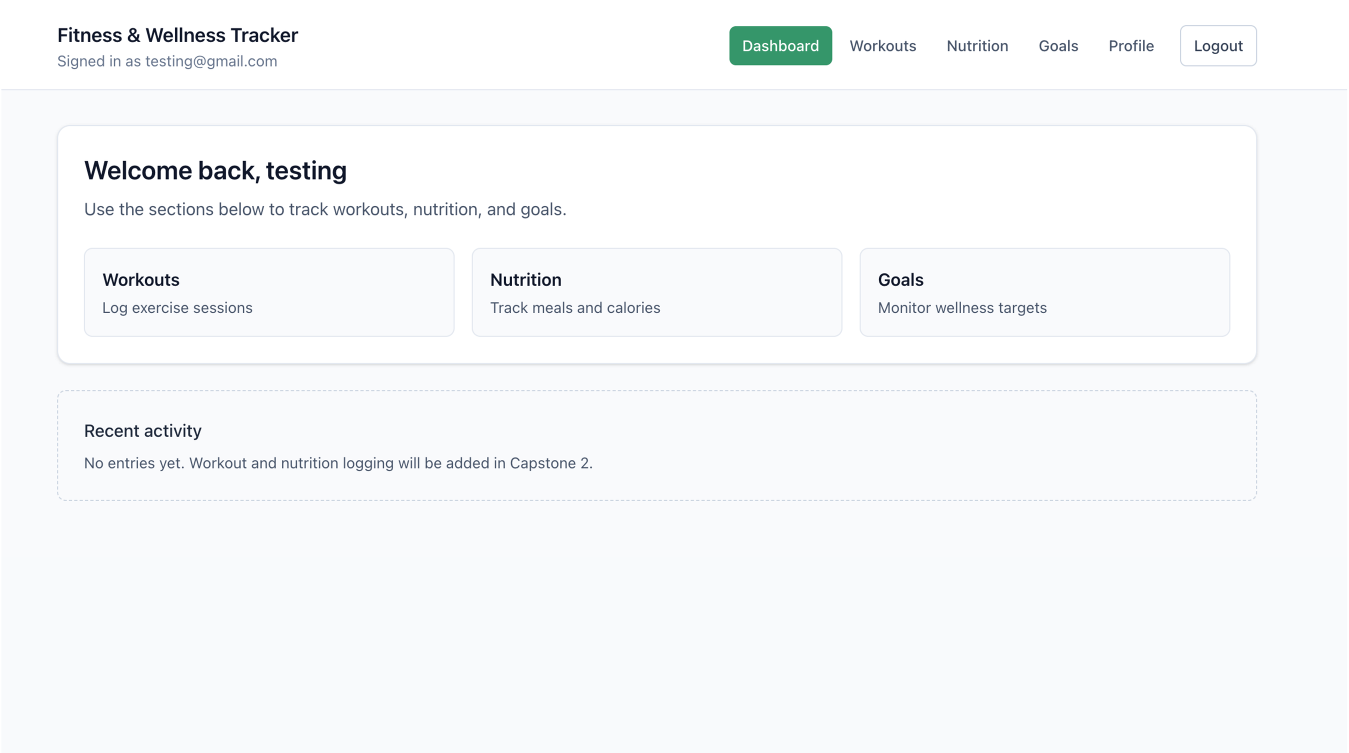
We designed a web application where users can create an account, log in securely, and use one dashboard to manage workouts, nutrition, and wellness goals. In Capstone 1 we finished the planning foundation (architecture, wireframes, database and API docs, and our scrum process) and built a React / Express / MongoDB starter with working authentication and a navigation shell so Capstone II can focus on feature logging.

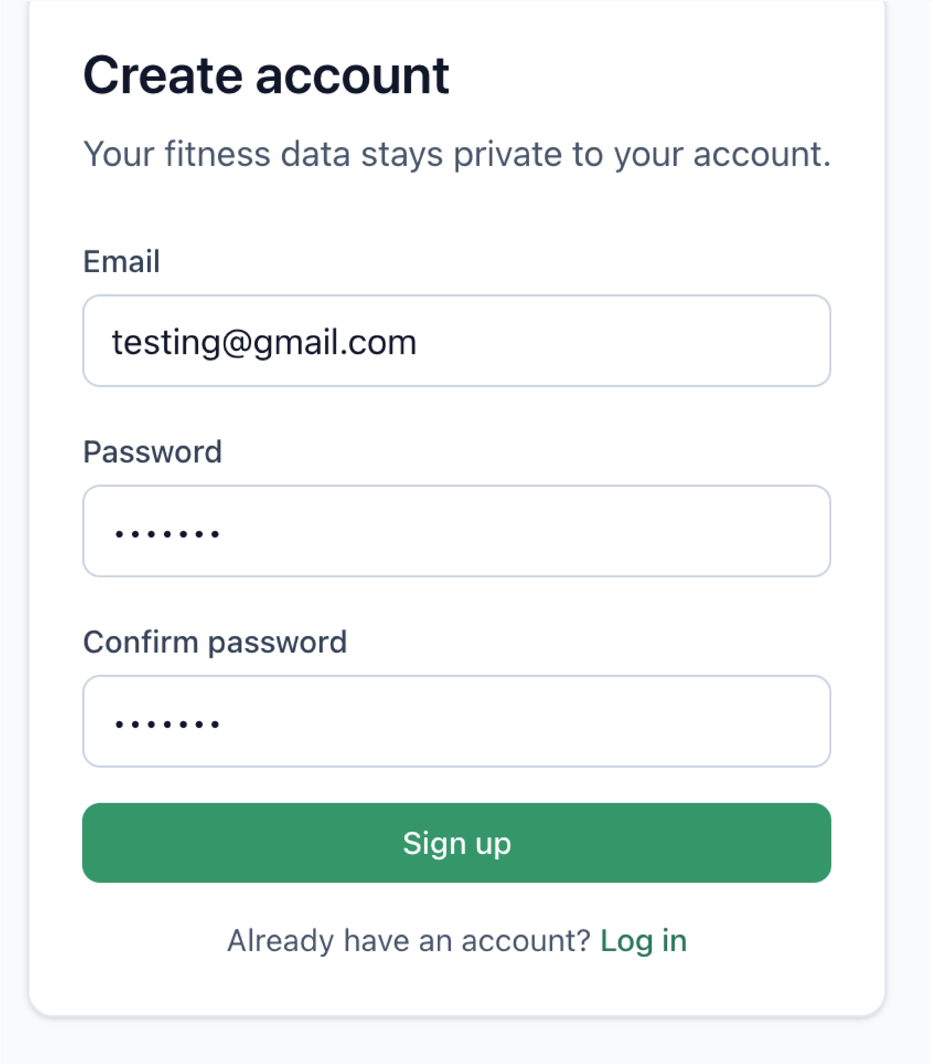
- One app for workouts, nutrition, and goals instead of separate tools
- JWT-based accounts so each user's fitness data stays private and scoped to their login
- Documented MVP roadmap and API contracts so we can implement features in Capstone II without restarting design

Architecture & How It Works

- The React UI calls JSON REST endpoints under /api; protected pages send a Bearer JWT after login.
- Express routes validate requests, apply auth middleware, and use Mongoose models for persistence.
- MongoDB stores users plus planned collections for workouts, nutrition logs, and goals, matching our ERD and schema docs.

Key Features & Results





- Dashboard navigation shell for Workouts, Nutrition, Goals, and Profile (feature pages ready for Capstone II logging)
- Mongoose models and REST contracts documented for workout, nutrition, and goal endpoints.
- Capstone I outcome: planning accepted through Sprint 4; Sprint 5 final documentation review in progress.
- Process: daily scrums, backlog grooming, sprint planning / review / retro, and a clear handoff for Capstone II.

The Team

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Technologies

- React, Vite, Tailwind CSS, React Router
- Node.js, Express, JWT
- MongoDB, Mongoose
- GitHub, Mermaid diagrams, Markdown docs

Acknowledgments

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